獨自升級Kafka

2025.11.27

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小明第一天上班拿到了
升級Kafka cluster的任務
他等不及的想好好探索這個卡夫卡

請問小明要怎麼知道現在的Kafka是哪一版呢？

聰明的小明當然知道去 bin 目錄底下找工具用，當然是挑名稱有 version 的來用

```
~/code/kafka_2.13-4.0.1
❯ ls bin
connect-distributed.sh            kafka-delegation-tokens.sh        kafka-replica-verification.sh
connect-mirror-maker.sh           kafka-delete-records.sh           kafka-run-class.sh
connect-plugin-path.sh            kafka-dump-log.sh                 kafka-server-start.sh
connect-standalone.sh             kafka-e2e-latency.sh              kafka-server-stop.sh
kafka-acls.sh                     kafka-features.sh                 kafka-share-groups.sh
kafka-broker-api-versions.sh      kafka-get-offsets.sh              kafka-storage.sh
kafka-client-metrics.sh           kafka-groups.sh                   kafka-streams-application-reset.sh
kafka-cluster.sh                  kafka-jmx.sh                      kafka-topics.sh
kafka-configs.sh                  kafka-leader-election.sh          kafka-transactions.sh
kafka-console-consumer.sh         kafka-log-dirs.sh                 kafka-verifiable-consumer.sh
kafka-console-producer.sh         kafka-metadata-quorum.sh          kafka-verifiable-producer.sh
kafka-console-share-consumer.sh   kafka-metadata-shell.sh           trogdor.sh
kafka-consumer-groups.sh          kafka-producer-perf-test.sh       windows
kafka-consumer-perf-test.sh       kafka-reassign-partitions.sh
```

再問Claude Opus 4.5確認一下

```
❯ ./bin/kafka-broker-api-versions.sh --bootstrap-server localhost:9092 --version
4.0.1
❯ ./bin/kafka-topics.sh --version
4.0.1
```

小明：看來還沒有到最新版4.1呢，我要來升級一下，好興奮啊

於是小明就被開除了
每天跟老婆還有兒子過著啃米餅的生活

方法 2：kafka-metadata.sh（KRaft 專用）

```bash
./bin/kafka-metadata.sh --bootstrap-server localhost:9092 --command "status"

# 或
./bin/kafka-metadata.sh --bootstrap-server localhost:9092 --command "features"
```

```
~/code/kafka_2.13-4.0.1
❯ ./bin/kafka-metadata.sh --bootstrap-server localhost:9092 --command "status"
zsh: no such file or directory: ./bin/kafka-metadata.sh

~/code/kafka_2.13-4.0.1
❯ ./bin/kafka-metadata.sh --bootstrap-server localhost:9092 --command "features"
zsh: no such file or directory: ./bin/kafka-metadata.sh
```

```
❯ ./bin/kafka-metadata-shell.sh --snapshot /tmp/kraft-combined-logs/__cluster_metadata-0/00000000000000000000.log --command "cat"

usage: kafka-metadata-shell [-h] [--snapshot SNAPSHOT] [command [command ...]]
kafka-metadata-shell: error: unrecognized arguments: '--command'
```

方法 3：kafka-features.sh

bash

```
./bin/kafka-features.sh --bootstrap-server localhost:9092 describe

# 輸出：
# Feature: metadata.version  SupportedMinVersion: 1  SupportedMaxVersion: 17  Final
#                                                                  ^^
#                                                 17 = 3.9-IV0，所以是 Kafka 3.9
```

```
~/code/kafka_2.13-4.0.1                                                    16:08:56
❯ ./bin/kafka-features.sh --bootstrap-server localhost:9092 describe
Feature: eligible.leader.replicas.version       SupportedMinVersion: 0  SupportedMaxVersion: 1  FinalizedVersionLevel: 1        Epoch: 4760
Feature: group.version  SupportedMinVersion: 0  SupportedMaxVersion: 1  FinalizedVersionLevel: 1        Epoch: 4760
Feature: kraft.version  SupportedMinVersion: 0  SupportedMaxVersion: 1  FinalizedVersionLevel: 1        Epoch: 4760
Feature: metadata.version       SupportedMinVersion: 3.3-IV3    SupportedMaxVersion: UNKNOWN 29 FinalizedVersionLevel: UNKNOWN 29       Epoch: 4760
Feature: share.version  SupportedMinVersion: 0  SupportedMaxVersion: 1  FinalizedVersionLevel: 1        Epoch: 4760
Feature: streams.version        SupportedMinVersion: 0  SupportedMaxVersion: 1  FinalizedVersionLevel: 1        Epoch: 4760
Feature: transaction.version    SupportedMinVersion: 0  SupportedMaxVersion: 2  FinalizedVersionLevel: 2        Epoch: 4760
```

UKNOWN 29是什麼
IV3又是什麼

UNKOWN: 舊的Tool看不懂啦

29 跟 IV3 請查看MetadataVersion的code

MetadataVersion

```
/**
 * This class contains the different Kafka versions.
 * Right now, we use them for upgrades - users can configure the version of the API brokers will use to communicate between themselves.
 * This is only for inter-broker communications - when communicating with clients, the client decides on the API version.
 * <br>
 * Note that the ID we initialize for each version is important.
 * We consider a version newer than another if it is lower in the enum list (to avoid depending on lexicographic order)
 * <br>
 * Since the api protocol may change more than once within the same release and to facilitate people deploying code from
 * trunk, we have the concept of internal versions (first introduced during the 1.0 development cycle). For example,
 * the first time we introduce a version change in a release, say 1.0, we will add a config value "1.0-IV0" and a
 * corresponding enum constant IBP_1_0-IV0. We will also add a config value "1.0" that will be mapped to the
 * latest internal version object, which is IBP_1_0-IV0. When we change the protocol a second time while developing
 * 1.0, we will add a new config value "1.0-IV1" and a corresponding enum constant IBP_1_0-IV1. We will change
 * the config value "1.0" to map to the latest internal version IBP_1_0-IV1. The config value of
 * "1.0-IV0" is still mapped to IBP_1_0-IV0. This way, if people are deploying from trunk, they can use
 * "1.0-IV0" and "1.0-IV1" to upgrade one internal version at a time. For most people who just want to use
 * released version, they can use "1.0" when upgrading to the 1.0 release.
 */
```

MetadataVersion

MetadataVersion

MetadataVersion

所以是我用的binary太舊了

改成4.1就看的清楚了

```
❯ ./bin/kafka-features.sh --bootstrap-server localhost:9092 describe
Feature: eligible.leader.replicas.version       SupportedMinVersion: 0  SupportedMaxVersion: 1  FinalizedVersionLevel: 1        Epoch: 55
Feature: group.version  SupportedMinVersion: 0  SupportedMaxVersion: 1  FinalizedVersionLevel: 1        Epoch: 55
Feature: kraft.version  SupportedMinVersion: 0  SupportedMaxVersion: 1  FinalizedVersionLevel: 1        Epoch: 55
Feature: metadata.version       SupportedMinVersion: 3.3-IV3    SupportedMaxVersion: 4.1-IV1    FinalizedVersionLevel: 4.1-IV1  Epoch: 55
Feature: share.version  SupportedMinVersion: 0  SupportedMaxVersion: 1  FinalizedVersionLevel: 0        Epoch: 55
Feature: transaction.version    SupportedMinVersion: 0  SupportedMaxVersion: 2  FinalizedVersionLevel: 2        Epoch: 55
```

Upgrading ZooKeeper-based clusters (3.9)

1. ~~Update server.properties on all brokers and add the following properties. CURRENT_KAFKA_VERSION refers to the version you are upgrading from. CURRENT_MESSAGE_FORMAT_VERSION refers to the message format version currently in use. If you have previously overridden the message format version, you should keep its current value. Alternatively, if you are upgrading from a version prior to 0.11.0.x, then CURRENT_MESSAGE_FORMAT_VERSION should be set to match CURRENT_KAFKA_VERSION.~~
    - ~~inter.broker.protocol.version=CURRENT_KAFKA_VERSION (e.g. `3.8`, `3.7`, etc.)~~
    - ~~log.message.format.version=CURRENT_MESSAGE_FORMAT_VERSION (See potential performance impact following the upgrade for the details on what this configuration does.)~~
2. **If you are upgrading from version 2.4.0** or above, and you have not overridden the message format, then you only need to override the inter-broker protocol version. However, before doing this, make sure ZooKeeper is upgraded to version 3.8.3 or higher.
    - inter.broker.protocol.version=CURRENT_KAFKA_VERSION (e.g. `3.8`, `3.7`, etc.)
3. Upgrade the brokers one at a time: shut down the broker, update the code, and **restart** it. Once you have done so, the brokers will be running the latest version and you can verify that the cluster's behavior and performance meets expectations. It is still possible to downgrade at this point if there are any problems.
4. Once the cluster's behavior and performance has been verified, bump the protocol version by editing `inter.broker.protocol.version`and setting it to `3.9`.
5. **Restart** the brokers one by one for the new protocol version to take effect. Once the brokers begin using the latest protocol version, it will no longer be possible to downgrade the cluster to an older version.
6. ~~If you have overridden the message format version as instructed above, then you need to do one more rolling restart to upgrade it to its latest version. Once all (or most) consumers have been upgraded to 0.11.0 or later, change log.message.format.version to 3.8 on each broker and restart them one by one. Note that the older Scala clients, which are no longer maintained, do not support the message format introduced in 0.11, so to avoid conversion costs (or to take advantage of exactly once semantics), the newer Java clients must be used.~~

Upgrading KRaft-based clusters (3.9)

1. Upgrade the brokers one at a time: shut down the broker, update the code, and **restart** it. Once you have done so, the brokers will be running the latest version and you can verify that the cluster's behavior and performance meets expectations.
2. Once the cluster's behavior and performance has been verified, bump the metadata.version by running `bin/kafka-features.sh upgrade --metadata 3.9`

Demo 3.8 -> 3.9

Start a 3.8 Kraft Kafka

```
export KAFKA_REPO=~/code/kafka_2.13-3.8.1
cd $KAFKA_REPO
rm -rf /tmp/kraft-combined-logs
KAFKA_CLUSTER_ID="$(./bin/kafka-storage.sh random-uuid)"
bin/kafka-storage.sh format -t $KAFKA_CLUSTER_ID -c config/kraft/server.properties
bin/kafka-server-start.sh config/kraft/server.properties
```

Create and write events into the topic

```
bin/kafka-topics.sh --create --topic test --bootstrap-server localhost:9092
bin/kafka-console-producer.sh --topic test --bootstrap-server localhost:9092
```

Consume the events

```
bin/kafka-console-consumer.sh --topic test --from-beginning --bootstrap-server localhost:9092
```

Check the version

```
./bin/kafka-features.sh --bootstrap-server localhost:9092 describe
```

Demo

Upgrade the brokers one at a time

1. shut down the 3.8.1 broker
2. update the code

   ```
   cd kafka_2.13-3.9.1
   ```

3. and restart it

   ```
   bin/kafka-server-start.sh config/kraft/server.properties
   ```

4. bump the metadata.version

   ```
   bin/kafka-features.sh --bootstrap-server localhost:9092 upgrade --metadata 3.9
   ```

如果大禮包有你不想要的禮物怎麼辦

KIP-1022: Formatting and Updating Features

- 4.0開始採用
- 很多功能都可以擁有自己獨立的feature，不用跟metadataVersion綁一起了
  - e.g,
  - New group coordinator: **group.version**
  - share group: **share.version**
- 想要全部一起升級，採用合適的default value：從 `--metadata` 改成 `--release-version`
- 可以查看dependencies

3.9

```
❯ ./bin/kafka-features.sh --bootstrap-server localhost:9092 describe
Feature: kraft.version  SupportedMinVersion: 0  SupportedMaxVersion: 1  FinalizedVersionLevel: 1  Epoch: 29
Feature: metadata.version       SupportedMinVersion: UNKNOWN 1  SupportedMaxVersion: 3.9-IV0    FinalizedVersionLevel: 3.9-IV0     Epoch: 29
```

4.0

```
❯ ./bin/kafka-features.sh --bootstrap-server localhost:9092 describe
Feature: eligible.leader.replicas.version       SupportedMinVersion: 0  SupportedMaxVersion: 1  FinalizedVersionLevel: 0        Epoch: 16
Feature: group.version  SupportedMinVersion: 0  SupportedMaxVersion: 1  FinalizedVersionLevel: 1        Epoch: 16
Feature: kraft.version  SupportedMinVersion: 0  SupportedMaxVersion: 1  FinalizedVersionLevel: 1        Epoch: 16
Feature: metadata.version       SupportedMinVersion: 3.3-IV3    SupportedMaxVersion: 4.0-IV3    FinalizedVersionLevel: 4.0-IV3  Epoch: 16
Feature: transaction.version    SupportedMinVersion: 0  SupportedMaxVersion: 2  FinalizedVersionLevel: 2        Epoch: 16
```

4.1

```
❯ ./bin/kafka-features.sh --bootstrap-server localhost:9092 describe
Feature: eligible.leader.replicas.version       SupportedMinVersion: 0  SupportedMaxVersion: 1  FinalizedVersionLevel: 1        Epoch: 17
Feature: group.version  SupportedMinVersion: 0  SupportedMaxVersion: 1  FinalizedVersionLevel: 1        Epoch: 17
Feature: kraft.version  SupportedMinVersion: 0  SupportedMaxVersion: 1  FinalizedVersionLevel: 1        Epoch: 17
Feature: metadata.version       SupportedMinVersion: 3.3-IV3    SupportedMaxVersion: 4.1-IV1    FinalizedVersionLevel: 4.1-IV1  Epoch: 17
Feature: share.version  SupportedMinVersion: 0  SupportedMaxVersion: 1  FinalizedVersionLevel: 0        Epoch: 17
Feature: transaction.version    SupportedMinVersion: 0  SupportedMaxVersion: 2  FinalizedVersionLevel: 2        Epoch: 17
```

4.2

```
❯ ./bin/kafka-features.sh --bootstrap-server localhost:9092 describe
Feature: eligible.leader.replicas.version       SupportedMinVersion: 0  SupportedMaxVersion: 1  FinalizedVersionLevel: 1        Epoch: 17
Feature: group.version  SupportedMinVersion: 0  SupportedMaxVersion: 1  FinalizedVersionLevel: 1        Epoch: 17
Feature: kraft.version  SupportedMinVersion: 0  SupportedMaxVersion: 1  FinalizedVersionLevel: 1        Epoch: 17
Feature: metadata.version       SupportedMinVersion: 3.3-IV3    SupportedMaxVersion: 4.2-IV1    FinalizedVersionLevel: 4.2-IV1  Epoch: 17
Feature: share.version  SupportedMinVersion: 0  SupportedMaxVersion: 1  FinalizedVersionLevel: 1        Epoch: 17
Feature: streams.version        SupportedMinVersion: 0  SupportedMaxVersion: 1  FinalizedVersionLevel: 1        Epoch: 17
Feature: transaction.version    SupportedMinVersion: 0  SupportedMaxVersion: 2  FinalizedVersionLevel: 2        Epoch: 17
```

```
~/code/kafka  trunk *2 ?1
❯ ./bin/kafka-features.sh --bootstrap-server localhost:9092 version-mapping
metadata.version=29 (4.2-IV1)
kraft.version=1
transaction.version=2
group.version=1
eligible.leader.replicas.version=1
share.version=1
streams.version=1

~/code/kafka  trunk *2 ?1
❯ ./bin/kafka-features.sh --bootstrap-server localhost:9092 version-mapping --release-version 4.1
metadata.version=27 (4.1)
kraft.version=1
transaction.version=2
group.version=1
eligible.leader.replicas.version=1
share.version=0
streams.version=0

~/code/kafka  trunk *2 ?1
❯ ./bin/kafka-features.sh --bootstrap-server localhost:9092 version-mapping --release-version 4.0
metadata.version=25 (4.0)
kraft.version=1
transaction.version=2
group.version=1
eligible.leader.replicas.version=0
share.version=0
streams.version=0
```

假設我要從4.1降到4.0，用4.0的binary來downgrade，可以嗎？

```
❯ cd kafka_2.13-4.0.1
❯ ./bin/kafka-features.sh --bootstrap-server localhost:9092 downgrade --release-version 4.0
```

如果用舊的binary執行downgrade

```
~/code/kafka_2.13-4.0.1                                                    17:22:29
❯ ./bin/kafka-features.sh --bootstrap-server localhost:9092 describe
Feature: eligible.leader.replicas.version       SupportedMinVersion: 0  SupportedMaxVersion: 1   FinalizedVersionLevel: 1        Epoch: 13535
Feature: group.version  SupportedMinVersion: 0  SupportedMaxVersion: 1  FinalizedVersionLevel: 1 Epoch: 13535
Feature: kraft.version  SupportedMinVersion: 0  SupportedMaxVersion: 1  FinalizedVersionLevel: 1 Epoch: 13535
Feature: metadata.version       SupportedMinVersion: 3.3-IV3    SupportedMaxVersion: UNKNOWN 29  FinalizedVersionLevel: UNKNOWN 29       Epoch: 13535
Feature: share.version  SupportedMinVersion: 0  SupportedMaxVersion: 1  FinalizedVersionLevel: 1 Epoch: 13535
Feature: streams.version        SupportedMinVersion: 0  SupportedMaxVersion: 1  FinalizedVersionLevel: 1 Epoch: 13535
Feature: transaction.version    SupportedMinVersion: 0  SupportedMaxVersion: 2  FinalizedVersionLevel: 2 Epoch: 13535

~/code/kafka_2.13-4.0.1                                                    17:23:07
❯ ./bin/kafka-features.sh --bootstrap-server localhost:9092 downgrade --release-version 4.0
eligible.leader.replicas.version was downgraded to 0.
group.version was downgraded to 1.
kraft.version was downgraded to 1.
metadata.version was downgraded to 25.
transaction.version was downgraded to 2.

~/code/kafka_2.13-4.0.1                                                    17:23:14
❯ ./bin/kafka-features.sh --bootstrap-server localhost:9092 describe
Feature: eligible.leader.replicas.version       SupportedMinVersion: 0  SupportedMaxVersion: 1  FinalizedVersionLevel: 0        Epoch: 13558
Feature: group.version  SupportedMinVersion: 0  SupportedMaxVersion: 1  FinalizedVersionLevel: 1        Epoch: 13558
Feature: kraft.version  SupportedMinVersion: 0  SupportedMaxVersion: 1  FinalizedVersionLevel: 1        Epoch: 13558
Feature: metadata.version       SupportedMinVersion: 3.3-IV3    SupportedMaxVersion: UNKNOWN 29 FinalizedVersionLevel: 4.0-IV3  Epoch: 13558
Feature: share.version  SupportedMinVersion: 0  SupportedMaxVersion: 1  FinalizedVersionLevel: 1        Epoch: 13558
Feature: streams.version        SupportedMinVersion: 0  SupportedMaxVersion: 1  FinalizedVersionLevel: 1        Epoch: 13558
Feature: transaction.version    SupportedMinVersion: 0  SupportedMaxVersion: 2  FinalizedVersionLevel: 2        Epoch: 13558
```

Upgrading KRaft-based clusters (4.1)

1. Upgrade the brokers one at a time: shut down the broker, update the code, and restart it. Once you have done so, the brokers will be running the latest version and you can verify that the cluster's behavior and performance meet expectations.
2. Once the cluster's behavior and performance have been verified, finalize the upgrade by running `bin/kafka-features.sh --bootstrap-server localhost:9092 upgrade --release-version 4.0`
3. Note that cluster metadata downgrade is not supported in this version since it has metadata changes. Every MetadataVersion has a boolean parameter that indicates if there are metadata changes (i.e. `IBP_4_0_IV1(23, "4.0", "IV1", true)` means this version has metadata changes). Given your current and target versions, a downgrade is only possible if there are no metadata changes in the versions between.

Upgrading KRaft-based clusters (3.9)

1. Upgrade the brokers one at a time: shut down the broker, update the code, and restart it. Once you have done so, the brokers will be running the latest version and you can verify that the cluster's behavior and performance meets expectations.
2. Once the cluster's behavior and performance **has** been verified, bump the metadata.version by running `bin/kafka-features.sh upgrade --metadata 3.9`
3. Note that cluster metadata downgrade is not supported in this software version. Every MetadataVersion after 3.2.x has a boolean parameter that indicates if there are metadata changes (i.e. `IBP_3_3_IV3(7, "3.3", "IV3", true)`means this version has metadata changes). Given your current and target versions, a downgrade is only possible if there are no metadata changes in the versions between.

Upgrading KRaft-based clusters (4.1)

1. Upgrade the brokers one at a time: shut down the broker, update the code, and restart it. Once you have done so, the brokers will be running the latest version and you can verify that the cluster's behavior and performance meet expectations.
2. Once the cluster's behavior and performance **have** been verified, finalize the upgrade by running `bin/kafka-features.sh --bootstrap-server localhost:9092 upgrade --release-version 4.0`
3. Note that cluster metadata downgrade is not supported in this version since it has metadata changes. Every MetadataVersion has a boolean parameter that indicates if there are metadata changes (i.e. `IBP_4_0_IV1(23, "4.0", "IV1", true)` means this version has metadata changes). Given your current and target versions, a downgrade is only possible if there are no metadata changes in the versions between.

小明覺得Kafka 4.0太可怕了，想降回3.9請問他辦得到嗎

```
./bin/kafka-features.sh --bootstrap-server localhost:9092 downgrade --release-version 3.9
```

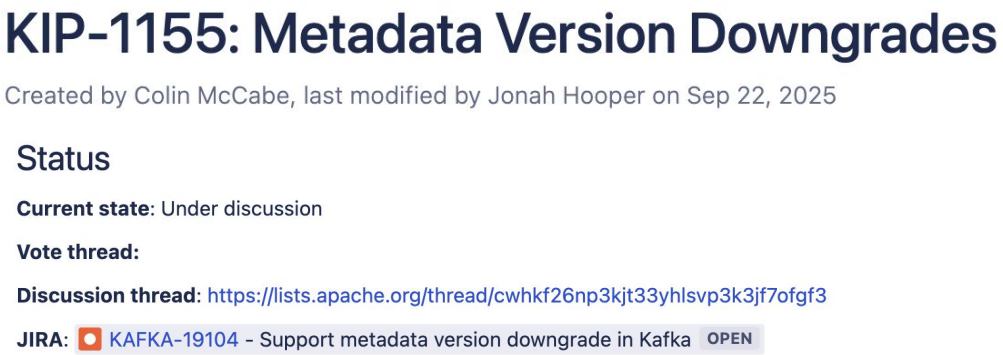
小明覺得Kafka 4.1太可怕了，想降回4.0請問他辦得到嗎

```
./bin/kafka-features.sh --bootstrap-server localhost:9092 downgrade --release-version 4.0
```

Note that cluster metadata downgrade is not supported in this software version. Every MetadataVersion after 3.2.x has a boolean parameter that indicates **if there are metadata changes** (i.e. `IBP_3_3_IV3(7, "3.3", "IV3", true)` means this version has metadata changes). Given your current and target versions, a downgrade is only possible if there are no metadata changes in the versions between.

Downgrade to older (changed) metadata

我想應該是沒機會了



KIP-1155: Metadata Version Downgrades

Created by Colin McCabe, last modified by Jonah Hooper on Sep 22, 2025

Status

Current state: Under discussion

Vote thread:

Discussion thread: https://lists.apache.org/thread/cwhkf26np3kjt33yhlsvp3k3jf7ofgf3

JIRA: KAFKA-19104 - Support metadata version downgrade in Kafka OPEN

https://cwiki.apache.org/confluence/display/KAFKA/KIP-1155%3A+Metadata+Version+Downgrades

Takeaway

- 怎麼查看broker版本
- Kraft升級少了一次改inter.broker.protocol.version config的重啟
- MetadataVersion以及他對應的IV是什麼，怎樣的情況可以降級
- 4.0新導入以後之後出現的各種feature是什麼東西

Reference

- https://medium.com/google-cloud/what-version-of-apache-kafka-are-you-running-is-a-hard-question-153976b14030